

Year 7 · Chapter 9 · The number plane: answers (teacher copy)

9.01 Grid references on maps (booklet p. 3)

Examples (model these live)

E1 What is at C2?

column C, row 2 · the Bank

E2 Give the grid reference of the Zoo.

column E, row 4 · E4

E3 Describe the move from the Bank to the Zoo.

C to E: 2 right · row 2 to 4: 2 up

Questions (19 on today's skill, 8 from earlier lessons)

1 Shop

2 B2

3 D2

4 A2

5 F3

6 Post

7 0.1^{7.09}

8 B3 to E2 · 3 right, 1 down

9 D → C, 3 → 1 · C1

10 874 mL^{8.10}

11 D → B, 4 → 1 · B1

12 F1 to C4 · 3 left, 3 up

13 94 cm^{8.02}

14 north: row 2 → 4 · east: D → E, so E4

15 $C = \pi \times 70 \approx 219.9$ cm^{8.03}

16 E → B, 3 → 1 · B1

17 north: row 3 → 4 · east: D → E, so E4

18 1 right, 1 down · 1 + 1 = 2 squares

19 50 cm²^{8.07}

20 D1; yes

21 rectangle: $12 \times 4 = 48$ · triangle: $\frac{1}{2} \times 12 \times 6 = 36$ · total: 84 m²^{8.08}

22 1 right, 3 up · 1 + 3 = 4 squares

23 C3; yes

24 5 ha^{8.04}

25 north: row 1 → 2 · east: B → D, so D2

26 549 cm = 5.49 m · 8 + 5.49 = 13.49 m^{8.01}

27 E → A, 4 → 3 · A3

9.02 Points in the first quadrant (booklet p. 5)

Examples (model these live)

E1 Write the coordinates of A.

across 4, up 2 · (4, 2)

E2 Which point is at (1, 5)?

across 1, up 5 · B

E3 A rectangle has corners (1, 1), (5, 1) and (5, 4). Find the fourth corner.

same x as (1, 1), same y as (5, 4) · (1, 4)

Questions (27 on today's skill, 10 from earlier lessons)

1 (5, 3)

2 C

3 2

4 (1, 5)

5 C

6 2

7 (0, 0)

8 2 ha^{8.04}

9 parallelogram

10 square

11 x: halfway between 0 and 2 is 1 · (1, 6)

12 E → C, 4 → 1 · C1^{9.01}

13 same x: count up · 9 - 2 = 7 units

14 x: halfway between 0 and 6 is 3 · (3, 2)

15 same x: count up · 5 - 0 = 5 units

16 F1^{9.01}

17 right-angled triangle

18 F → B, 5 → 3 · B3^{9.01}

19 fourth corner: (3, 4) · 4 × 4 = 16 square units

20 x: halfway between 2 and 4 is 3 · (3, 6)

21 fourth corner: (0, 6) · 2 × 4 = 8 square units

22 (4, 7) and (7, 7)

23 E → A, 4 → 2 · A2^{9.01}

24 x comes first: it is (1, 3).

25 60 000 L^{8.01}

26 fourth corner: (1, 4) · 2 × 3 = 6 square units

27 (2, 8) and (6, 8)

28 B4^{9.01}

29 x comes first: it is (1, 2).

30 (3, 5) and (5, 5)

31 fourth corner: (0, 4) · 3 × 3 = 9 square units

32 $54.25 \div 5 = \$10.85$ ^{7.07}

33 same x: count up · 5 - 3 = 2 units

34 A5^{9.01}

35 (2, 4) and (5, 4)

36 fourth corner: (1, 7) · 4 × 4 = 16 square units

37 E → C, 5 → 1 · C1^{9.01}

9.03 The number plane (booklet p. 7)

Examples (model these live)

E1 Write the coordinates of A.

3 left, 2 down · (-3, -2)

E2 Which quadrant is (-4, 1) in?

x negative, y positive · 2nd quadrant

E3 Reflect (3, -2) in the x-axis.

the sign of y changes · (3, 2)

Questions (31 on today's skill, 12 from earlier lessons)

1 (-3, -2)

2 3rd quadrant

3 A

4 (-4, 4)

5 3rd quadrant

6 1st quadrant

7 C

8 (2, 1)

9 0.16^{7.09}

10 the sign of x changes · (3, 8)

11 4th quadrant

12 the sign of x changes · (-4, 6)

13 x: 1 - 1 = 0, y: 5 + 3 = 8 · (0, 8)

14 C → A, 3 → 2 · A2^{9.01}

15 1 to the y-axis, 1 more · 2 units

16 the sign of x changes · (-2, 6)

17 $36 - 3 = 33$ cm²^{8.08}

18 x: $-3 + 1 = -2$, y: $-2 - 3 = -5$ · (-2, -5)

19 x: $5 - 6 = -1$, y: $2 - 5 = -3$ · (-1, -3)

20 1 to the y-axis, 1 more · 2 units

21 6^{9.02}

22 the sign of y changes · (7, -2)

23 -20, -10^{1.02}

24 both signs change: (-3, -3) · 3rd quadrant

25 x: $-2 - 6 = -8$, y: $4 + 2 = 6$ · (-8, 6)

26 $A = \frac{1}{2} \times 5.6 \times 10 = 28$ m²^{8.06}

27 6 to the y-axis, 6 more · 12 units

28 E5^{9.01}

29 both signs change: (-6, 2) · 2nd quadrant

30 same x as the first, same y as the third · (-1, -2)

31 parallelogram^{9.02}

32 y-axis: (-1, 5) · x-axis: (-1, -5)

33 147 m³^{8.11}

34 both signs change: (5, -6) · 4th quadrant

35 same x as the first, same y as the third · (-1, 1)

36 same x as the first, same y as the third · (-3, 5)

37 y-axis: (-4, 5) · x-axis: (-4, -5)

38 $3 \times 11 = 33$ cm^{8.02}

39 both signs change: (-2, -3) · 3rd quadrant

40 A^{9.02}

41 4 to the y-axis, 4 more · 8 units

42 same x as the first, same y as the third · (-1, -1)

43 -8^{1.02}

9.04 Plotting tables of values (booklet p. 9)

Examples (model these live)

E1 Write the points from the table.
one point for each column · (0, 1), (1, 3), (2, 5), (3, 7)

E2 Describe the pattern in the table.
y goes up by 2 each time · y starts at 1

E3 Plot the points. Do they lie in a straight line?
plot each point · yes

Questions (30 on today's skill, 9 from earlier lessons)

- | | | | |
|--|--|---|---|
| 1 (0, 4), (1, 7), (2, 10), (3, 13) | 13 y goes up by 5 each time · y starts at 9 | 22 6 to the y-axis, 6 more · 12 units ^{9.03} | 31 starts at 4, up 5 each time · $4 + 10 \times 5 = 54$ |
| 2 13 | 14 y goes up by $4 \cdot 11$ | 23 y goes up by $5 \cdot 15$ | 32 up 3 each time: $3x$, then + $4 \cdot y = 3x + 4$ |
| 3 (0, 1), (1, 4), (2, 7), (3, 10) | 15 $D \rightarrow C, 4 \rightarrow 1 \cdot C1$ ^{9.01} | 24 y goes down by 2 each time · y starts at 8 | 33 up 3 each time: $3x$, then + $1 \cdot y = 3x + 1$ |
| 4 9 | 16 No: y goes up by 1, 3, 5 (not the same each time). | 25 $= \frac{54}{100} = 0.54$ ^{7.02} | 34 plot and join the points · at (0, 3) |
| 5 (0, 2), (1, 3), (2, 4), (3, 5) | 17 y goes up by 4 each time · y starts at 5 | 26 starts at 2, up 4 each time · $2 + 10 \times 4 = 42$ | 35 $\times 100: 12 \div 6 \cdot = 2$ ^{7.08} |
| 6 5 | 18 y goes up by $4 \cdot 13$ | 27 up 5 each time: $5x$, then + $1 \cdot y = 5x + 1$ | 36 starts at 5, up 4 each time · $5 + 10 \times 4 = 45$ |
| 7 12 | 19 y goes up by 2 each time · y starts at 9 | 28 rectangle ^{9.02} | 37 A ^{9.02} |
| 8 (0, 0), (1, 2), (2, 4), (3, 6) | 20 8 ^{9.02} | 29 plot and join the points · at (0, 4) | 38 up 2 each time: $2x$, then + $2 \cdot y = 2x + 2$ |
| 9 y goes up by 4 each time · y starts at 8 | 21 starts at 6, up 5 each time · $6 + 10 \times 5 = 56$ | 30 8 ^{8.09} | 39 plot and join the points · at (0, 4) |
| 10 9 000 L ^{8.01} | | | |
| 11 6 | | | |
| 12 (0, 3), (1, 5), (2, 7), (3, 9) | | | |

9.05 Tables of values from rules (booklet p. 11)

Examples (model these live)

E1 Find y when $x = 4$, for $y = 3x - 2$.
 $y = 3 \times 4 - 2 \cdot = 10$

E2 Complete the table for $y = 2x + 1$.
substitute $x = 0, 1, 2, 3 \cdot y: 1, 3, 5, 7$

E3 Find x when $y = 13$, for $y = 2x + 3$.
 $2x + 3 = 13 \cdot x = 5$

Questions (34 on today's skill, 11 from earlier lessons)

- | | | | |
|--|--|---|--|
| 1 29 | 14 19 | 23 \$2.55 ^{7.11} | 34 $2x + 4 = 14 \cdot x = 5$ |
| 2 5, 8, 11 | 15 $x = -2: 2 \times (-2) + 1 = -3 \cdot -3, -1, 1, 3$ | 24 $x = -2: 3 \times (-2) + 1 = -5 \cdot -5, -2, 1, 4$ | 35 $4 \times 2 + 1 = 9 \cdot$ no: $9 \neq 8$ |
| 3 15 | 16 $x = -2: 2 \times (-2) + 2 = -2 \cdot -2, 0, 2, 4$ | 25 D3 ^{9.01} | 36 $D \rightarrow C, 3 \rightarrow 2 \cdot C2$ ^{9.01} |
| 4 5, 9, 13 | 17 test $x = 1, 2$ and 3 in each · $y = 3x + 5$ | 26 $2x - 2 = 14 \cdot x = 8$ | 37 substitute $k = 0, 1, 2, 3 \cdot$ \$3, \$8, \$13, \$18 |
| 5 -1 | 18 $E \rightarrow B, 4 \rightarrow 2 \cdot B2$ ^{9.01} | 27 test $x = 1, 2$ and 3 in each · $y = 3x + 1$ | 38 $3 \times 3 + 1 = 10 \cdot$ yes |
| 6 3, 7, 11 | 19 y goes down by 1 · 6, 5, 4, 3 | 28 A ^{9.02} | 39 substitute $k = 0, 1, 2, 3 \cdot$ \$6, \$10, \$14, \$18 |
| 7 6 | 20 $x = -2: 4 \times (-2) + 2 = -6 \cdot -6, -2, 2, 6$ | 29 $5x - 2 = 13 \cdot x = 3$ | 40 6 ^{8.09} |
| 8 4, 8, 12 | 21 test $x = 1, 2$ and 3 in each · $y = 2x + 4$ | 30 $2 \times 1 + 4 = 6 \cdot$ yes | 41 $2x - 1 = 5 \cdot x = 3$ |
| 9 0, 2, 4 | 22 y goes down by 2 · 5, 3, 1, -1 | 31 5 L = 5 000 mL · 5 000 ÷ 250 = 20 ^{8.13} | 42 F5 ^{9.01} |
| 10 20 | | 32 substitute $k = 0, 1, 2, 3 \cdot$ \$3, \$9, \$15, \$21 | 43 y goes down by 3 · 9, 6, 3, 0 |
| 11 $x = -2: 2 \times (-2) - 3 = -7 \cdot -7, -5, -3, -1$ | | 33 4 ^{5.05} | 44 $2 \times 1 + 5 = 7 \cdot$ no: $7 \neq 8$ |
| 12 81 cm ² ^{8.05} | | | 45 6 ^{9.02} |
| 13 1, 4, 7 | | | |

9.06 Graphing linear equations (booklet p. 13)

Examples (model these live)

E1 Graph $y = 2x - 1$ for $x = 0$ to 3.
table: (0, -1), (1, 1), (2, 3), (3, 5) · plot and rule a line

E2 Does (3, 10) lie on $y = 3x + 1$?
 $3 \times 3 + 1 = 10 \cdot$ yes

E3 Where does $y = x + 4$ cross the y-axis?
 $x = 0: y = 4 \cdot (0, 4)$

Questions (32 on today's skill, 11 from earlier lessons)

- | | | | |
|---------------------------|--|--|---|
| 1 no | 12 (0, -1), (1, 0), (2, 1) · plot and rule a line | 21 (0, 1), (1, 2), (2, 3) · plot and rule a line | 28 starts at 4, up 4 for each 1 across · $y = 4x + 4$ |
| 2 (0, -2) | 13 68 cm ^{8.02} | 22 165 cm ² ^{8.07} | 29 $A = \frac{1}{2} \times 6.8 \times 10 \cdot = 34$ m ² ^{8.06} |
| 3 yes | 14 (0, 3) | 23 make a table for each: find the same point · (1, 5) | 30 $y = 6x - 2: y$ goes up by 6 each step, not 3. |
| 4 (0, -4) | 15 yes | 24 B ^{9.02} | 31 (0, 1), (1, 4), (2, 7), (3, 10) ^{9.04} |
| 5 no | 16 (0, 0), (1, 1), (2, 2) · plot and rule a line | 25 $3 \times (-1) - 2 = -5 \cdot$ yes | 32 make a table for each: find the same point · (2, 2) |
| 6 (0, 4) | 17 $2 \times (-2) + 5 = 1 \cdot$ yes | 26 make a table for each: find the same point · (1, 2) | 33 starts at 1, up 3 for each 1 across · $y = 3x + 1$ |
| 7 yes | 18 $C \rightarrow B, 3 \rightarrow 1 \cdot B1$ ^{9.01} | 27 make a table for each: find the same point · (2, 3) | |
| 8 (0, 3) | 19 (0, 1), (1, 3), (2, 5) · plot and rule a line | | |
| 9 5, 6, 7 ^{9.05} | 20 $4 \times 0 + 1 = 1 \cdot$ no: $1 \neq 3$ | | |
| 10 (0, 1) | | | |
| 11 no | | | |

34 E3 to C1 · 2 left, 2 down
9.01

35 starts at 5, up 2 for each
1 across · $y = 2x + 5$

36 $y = 5x - 2$: y goes up by 5
each step, not 3.

37 32 m 8.03

38 make a table for each:
find the same point · (2, 4)

39 3 9.02

40 $3 \times 0 - 2 = -2$ · no: $-2 \neq 0$

41 starts at 1, up 5 for each
1 across · $y = 5x + 1$

42 (-3, 3) 9.03

43 $y = 6x - 2$: y goes up by 6
each step, not 4.